Computer Vision / EE6102
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20260038

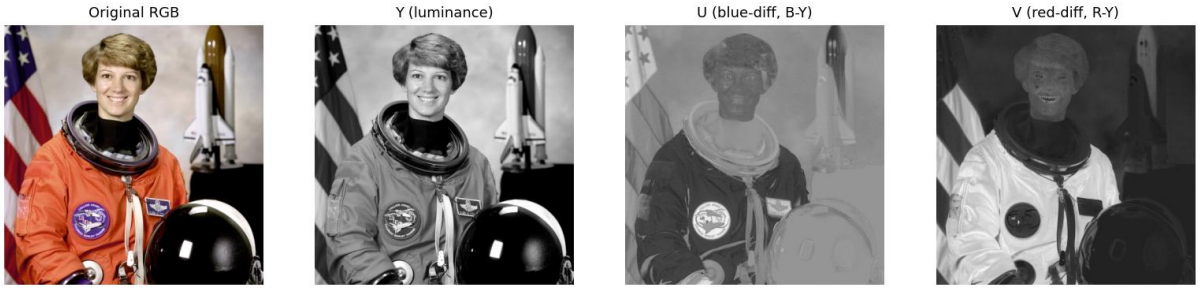
Homework 1 - Computer Vision

homework1_cv.py for the code. Generated figures are saved in the output/ folder.

Part A. Point Processing and Histogram-Based Image Enhancement

A-1. RGB -> YUV Conversion

The given matrix is applied directly to separate luminance Y from chrominance U, V. Since R,G,B in [0,255] and the first row sums to 1, Y lies in [0,255] (luminance). U and V are color differences and can be negative, so they are kept as floats without clipping or 8-bit casting.



Y range = [0.00, 255.00], U range = [-58.34, 41.36], V range = [-14.72, 93.85]

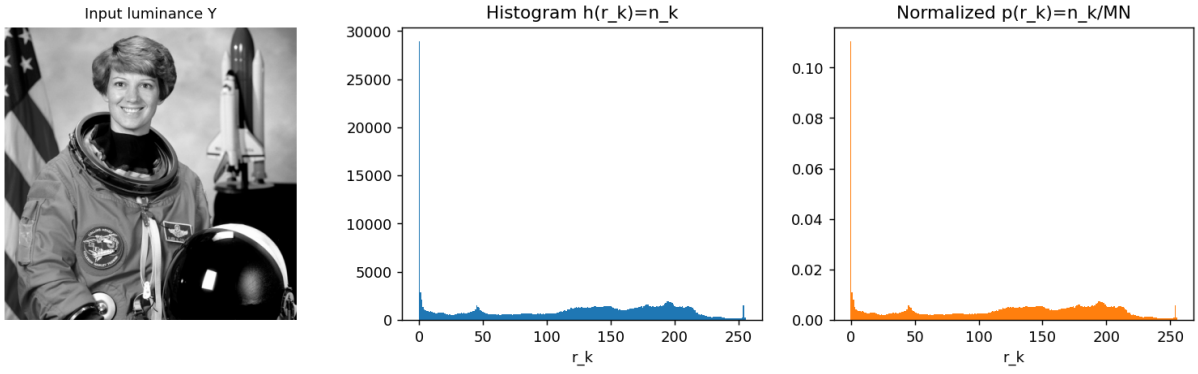
corr(U, B-Y) = 1.000, corr(V, R-Y) = 1.000

Observed in the figure: the Y image clearly shows the face, suit, helmet and the flag behind. The orange suit is bright in V (red-difference) and dark in U (blue-difference), since orange has high R and low B.

What do Y, U and V represent?: the weighted sum reflecting human brightness sensitivity (G has the largest weight, 0.587). It carries most of the structure and edges. U and V are approximately proportional to B-Y and R-Y, respectively. The supplied matrix coefficients are rounded, so correlations displayed as 1.000 do not prove exact proportionality.

A-2. Histogram Computation

Y is quantized to 8-bit (r_k in {0,...,255}) and counted per level directly: h(r_k) = n_k, with normalized histogram p(r_k) = n_k / MN.



sum(h) = 262144 (= M*N), sum(p) = 1.0000

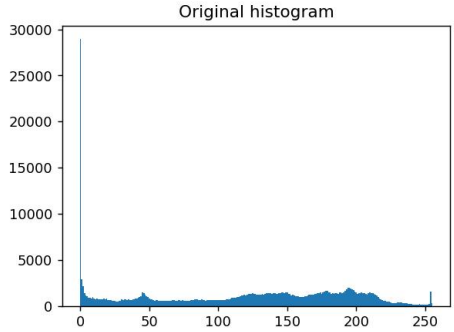
in the loop, k is the intensity r_k and np.sum(q==k) is the pixel count n_k.

the normalized histogram has the same shape as the raw one because normalization divides every bar by the same constant MN; only the y-axis scale changes (counts vs. probability, sum = 1). A distribution concentrated in a narrow band means low contrast.

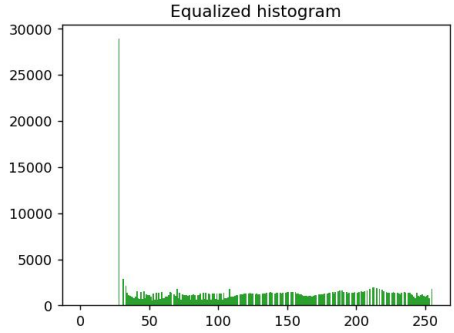
A-3. Histogram Equalization (HE)

The CDF, CDF(r_k) = sum_{j<=k} p(r_j), is used as the transform: s_k = (L-1)*CDF(r_k). This globally spreads the distribution toward uniform, increasing contrast. The equalized Y is recombined with the original U, V and converted back to RGB.



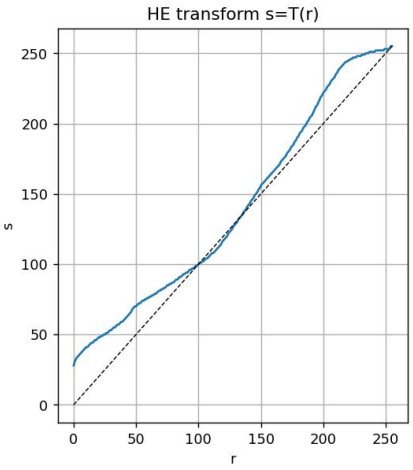












Y mean 115.4 -> 129.5, std 75.1 -> 71.4

Observed in the figure: the HE result is brighter overall, and dark helmet shadows and the background flag/details become more visible; the transform curve rising above the diagonal reflects this lifting of dark values.

the transform slope is dT/dr = (L-1)*p(r): where the histogram is dense (large p), the slope is steep and those intensities are spread apart, increasing contrast there.

the equalized histogram becomes comb-like for two reasons: dense regions skip output levels (empty-level gaps), while sparse regions - whose CDF rises slowly - round several inputs to the same output (many-to-one merging).

The mean increases from 115.4 to 129.5, whereas the global standard deviation decreases from 75.1 to 71.4. Increased brightness therefore does not imply increased global contrast. Black input maps to approximately 28, a dark gray, and contrast changes differently across populated and sparse intensity intervals.

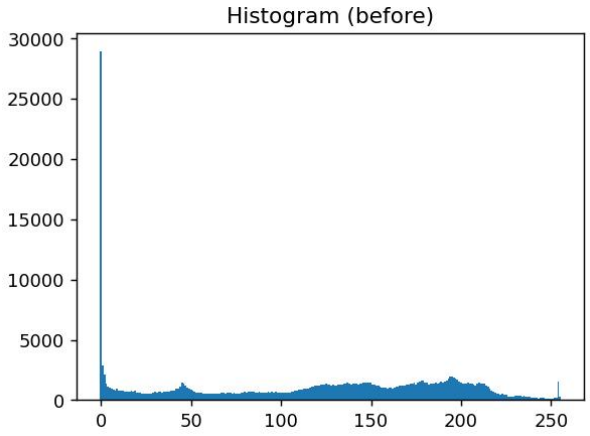
A-4. Contrast Stretching

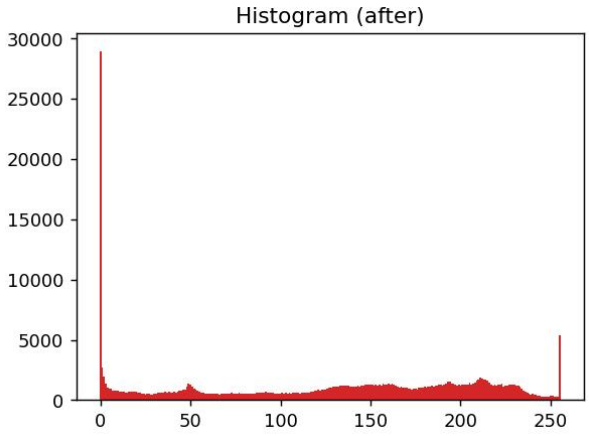
The percentile levels are the first levels at which the histogram CDF of quantized Y reaches 0.02 and 0.98. They are r_min=0 and r_max=235 for this input.

A piecewise-linear transform maps [r_min, r_max] linearly onto [0, L-1]. r_min, r_max are the 2nd/98th percentiles (robust to outliers).









r_min(2%) = 0.0, r_max(98%) = 235.0, gain = (L-1)/(r_max-r_min) = 1.085

Observed in the figure: the original and stretched images are visually almost indistinguishable and the before/after histograms are nearly identical - the dynamic range is already full.

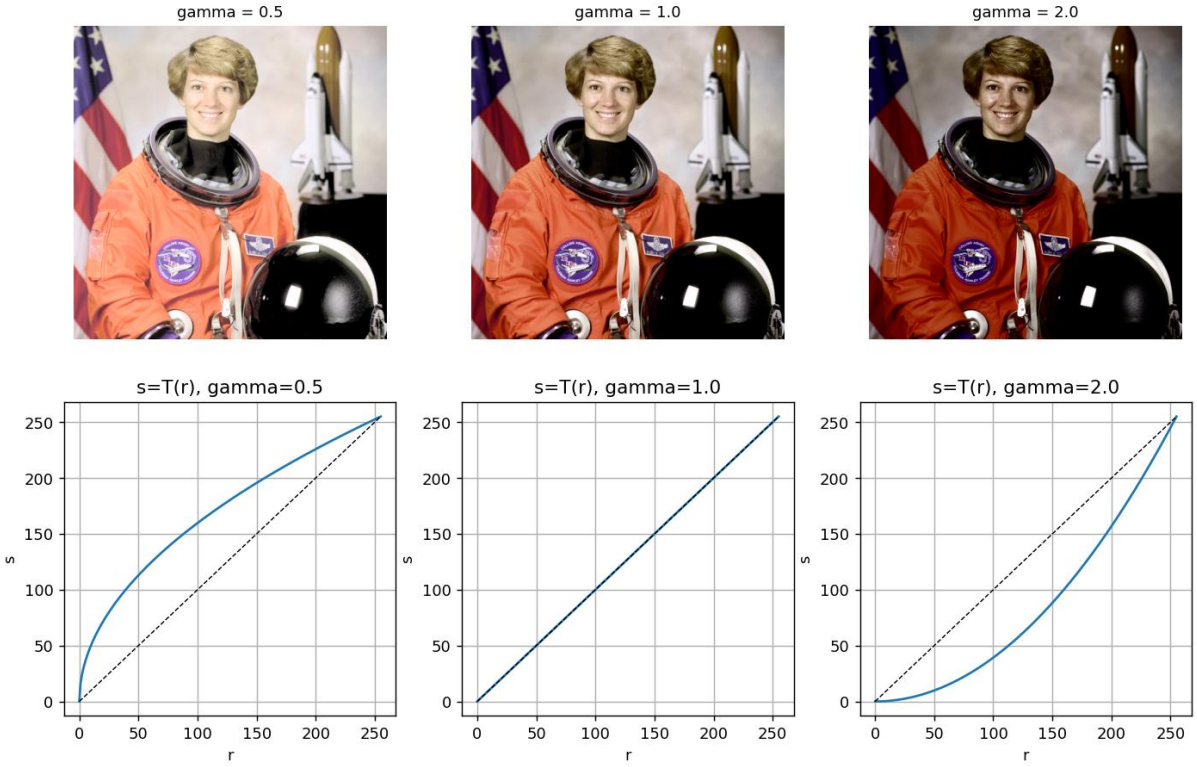
this image already spans nearly the full range (gain ~= 1.09), so the before/after difference is small. Stretching is most effective on low-dynamic-range (hazy) images.

How does contrast stretching differ from HE? Stretching applies a constant linear gain inside the chosen central range but clips the tails. Histogram equalization is a nonlinear CDF mapping with different gains across intensity intervals.

A-5. Gamma Correction (power-law)

Mean luminance: original 115.406; gamma=0.5 153.380; gamma=1 115.406; gamma=2 74.358.

Normalize r_n=r/255, apply s_n=c*r_n^gamma with c=1, and convert back using s=255*s_n. The tested gamma values are 0.5, 1.0 and 2.0. In original intensity units, ds/dr=gamma*(r/255)^(gamma-1).



Observed in the figure: at gamma=0.5 the dark helmet/shadows brighten and reveal detail; at gamma=2.0 the whole image darkens and the bright face/background are compressed.

gamma < 1 (0.5): brightens and expands the dark tones. In normalized coordinates, the slope gamma*r_n^(gamma-1) tends to infinity as r->0, so dark values are pulled apart (higher contrast), revealing shadow detail; the bright region is compressed.

gamma = 1: identity transform (unchanged).

gamma > 1 (2.0): darkens. Here the slope tends to 0 as r->0, so dark tones are compressed (shadows crushed) while bright tones expand.

A-6. Adaptive Histogram Equalization (AHE)

Mean standard deviation across 64x64 tiles: original 50.052; HE 48.098; AHE 66.884. This includes structure, contrast and noise.

The image is divided into 8x8 tiles and HE is applied independently in each tile (no interpolation).



Observed in the figure: the AHE result shows clear blocking along tile boundaries and blotchy noise in the dark background and smooth suit regions, while local contrast is much stronger than global HE.

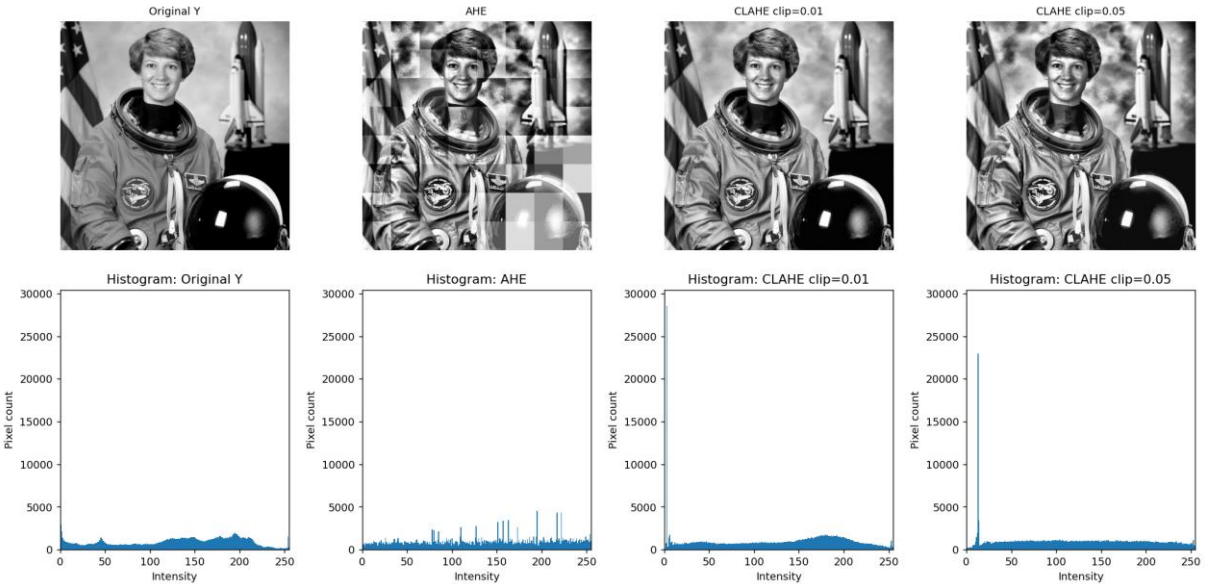
Why does AHE improve local contrast better than global HE? using a local CDF per tile improves local contrast far more than global HE.

Why can AHE amplify noise? in nearly uniform tiles (e.g., dark background, smooth parts of the suit) the narrow brightness range is force-stretched to 0..255, so faint noise shows up as blotches. Tile-boundary discontinuities also cause blocking artifacts - both addressed by CLAHE.

A-7. CLAHE (Contrast Limited AHE)

Mean tile standard deviation: AHE 66.884; CLAHE clip=0.01 59.580; clip=0.05 65.434. This is not a noise-only measurement.

Each tile histogram is clipped at a clip limit, the excess is redistributed uniformly, a local CDF forms the mapping, and bilinear interpolation between tile centers removes boundary artifacts. clip = 0.01 and 0.05 are compared.



Tile 64x64 = 4096 px: clip = 0.01 -> clip_count = 40, clip = 0.05 -> clip_count = 204

Observed in the figure: clip=0.01 reduces boundary discontinuities and excessive dark-region changes relative to independent-tile AHE; clip=0.05 produces stronger local contrast and background variation. Histograms of the original, AHE and both CLAHE outputs use the same count scale. A cap on local input histograms does not cap peaks in the global output histogram.

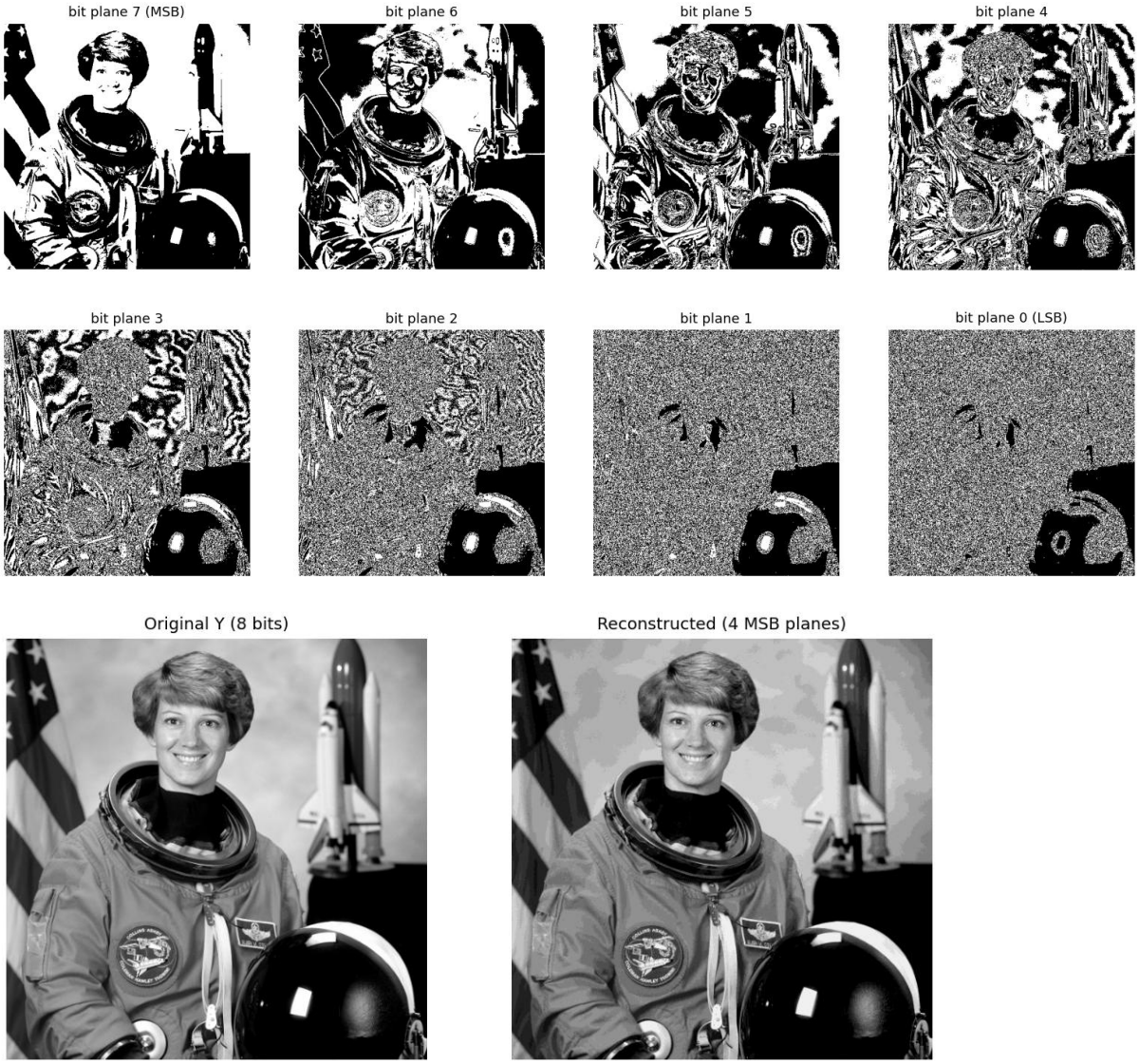
How does CLAHE reduce noise amplification? clipping reduces concentration in each local histogram and moderates steep local CDF mappings. Interpolation reduces discontinuities across tile boundaries. These are different mechanisms; output appearance alone is not a direct measurement of noise.

How does the clip limit affect local contrast? clip_count=floor(clip*tile_pixel_count), giving 40 or 204. Excess E is redistributed once across all 256 bins, so h_final=min(h,clip_count)+E/256. The configured count is therefore not a strict upper bound after redistribution. Before LUT rounding, the mapping increment is 255*h_final/4096.

What happens if the clip limit is too small or too large? A small limit makes the local histogram more uniform and the mapping closer to identity, reducing enhancement. A sufficiently large limit removes clipping and approaches interpolated AHE. The AHE baseline here has no interpolation, so it need not become exactly the same image.

A-8. Bit-Plane Slicing

Y(x,y) = sum_k b_k * 2^k is decomposed into 8 bit planes. The image is reconstructed from the 4 most significant planes (bit 4..7) and compared with the original.



4-MSB reconstruction: MSE = 67.19, PSNR = 29.86 dB

Observed in the figure: bit 7-5 alone already identify the face/suit/helmet and bit 4 adds texture; from bit 3 down the planes become increasingly random, and bit 0 (LSB) shows fine variation, which cannot be classified as pure noise from this display alone.

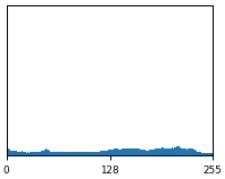
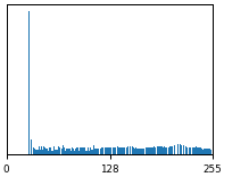
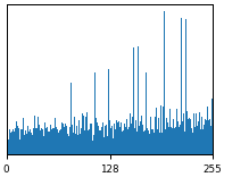
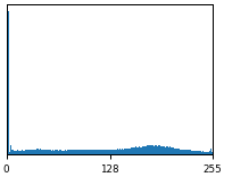
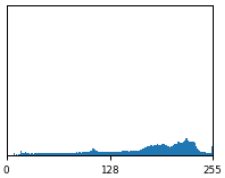
Which bit planes contain most of the visual structure? high-order planes (bit 7/6/5) carry large brightness changes - most of the visual structure/outline.

Which bit planes mainly contain fine details or noise? Low-order planes (bit 1/0) have tiny weight and resemble fine texture/noise.

two reasons the 4 MSB are the most important: (1) their weights 16+32+64+128 = 240 are about 94% of the max value 255; (2) the reconstruction PSNR ~= 29.9 dB and is visually almost identical. Dropping the low bits causes faint false contours in smooth gradients.

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A-9. Comparison and Discussion

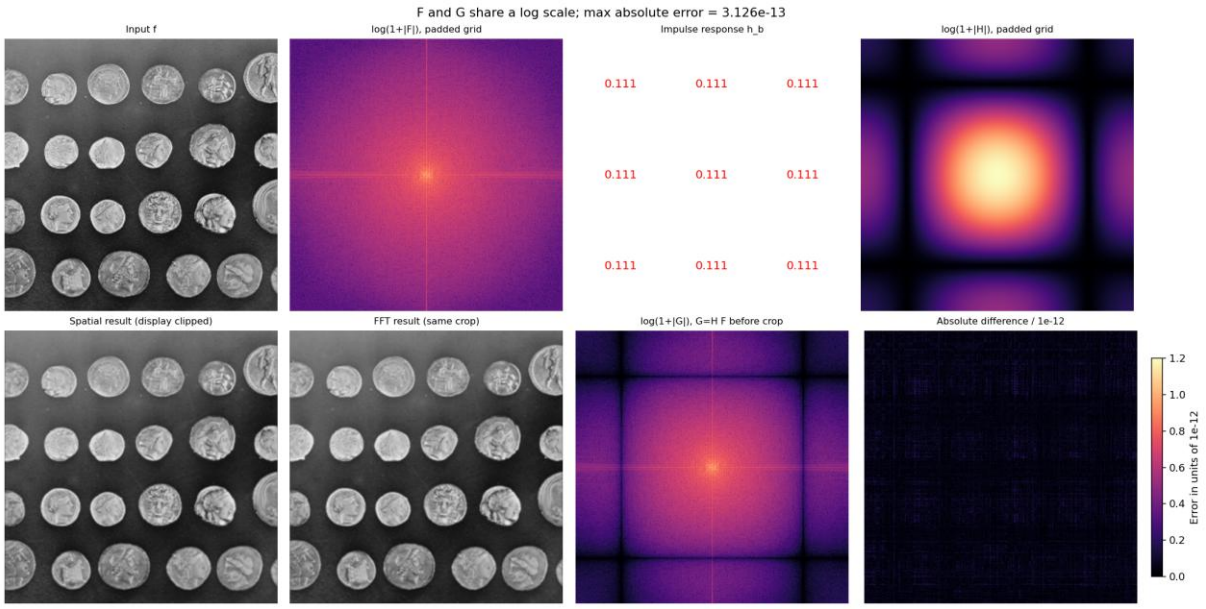
Method	Result image	Scope	Contrast improvement	Noise sensitivity	Histogram	Complexity	Suitable applications
Contrast stretching		Global	Constant gain within the chosen range; tails clipped.	Signal and noise both increase where gain is large.		O(LMN+L)	Narrow dynamic range; preprocessing
Histogram equalization		Global	CDF-dependent gain across intensity intervals.	Can amplify noise in populated intervals; higher mean does not imply higher global contrast.		O(LMN+L)	Redistribute a global intensity distribution
AHE		Local	Strong local enhancement using tile CDFs.	Amplifies small variations in flat tiles; no interpolation here, so seams can appear.		O(LMN+TL)	Spatially varying contrast; consider noise and seams
CLAHE		Local	Clipping moderates local gain; interpolation reduces seams.	Larger clip limits can increase detail and noise. Redistribution means the count cap is not strict.		O(LMN+TL)	Controlled local enhancement; low-light images
Gamma correction		Global	Gamma < 1 expands dark tones; gamma > 1 expands bright tones.	Existing noise can increase in the expanded intensity range.		O(MN)	Gamma and tone adjustment

Complexity describes this implementation, which scans pixels for each intensity level: image MxN, L=256, T=64. With L and T fixed, cost is linear in pixel count. Single-pass histogram counting can use O(MN+L). Displayed examples: CLAHE clip=0.01 and gamma=0.5.

Part B. Spatial- and Frequency-Domain Filtering

B-1. Blur Filtering

Top row shows the four required displays: (1) input f, (2) log|F|, (3) impulse response h_b, (4) log|H_b|. Bottom row shows the spatial result g_s = h_b*f, the frequency result g_f = F^-1{H_b F}, the output spectrum log(1+|G_f|), and the pixelwise difference |g_s - g_f|.



spatial vs frequency: MSE = 3.61e-27, PSNR = 312.6 dB, SSIM = 1.000000

Observed in the figure: coin edges and texture are softened equally in both outputs. The difference map displays error in units of 1e-12, so round-off patterns remain visible. Its maximum, approximately 3.13e-13, is negligible relative to the intensity range 0 to 255.

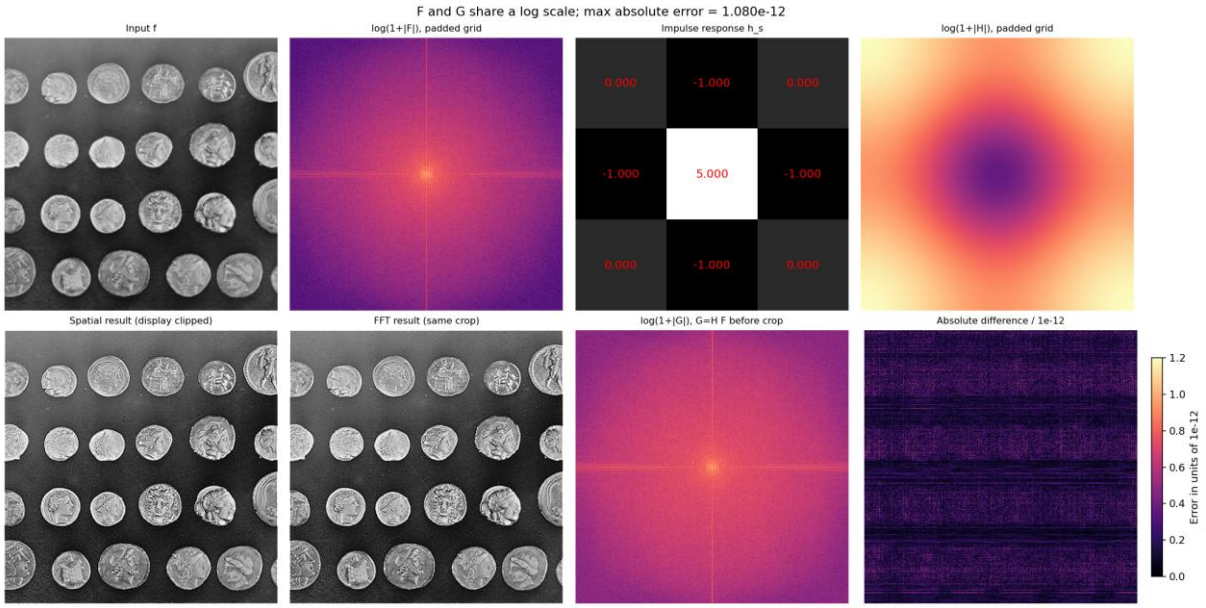
Reading the spectra: F, H and G come from the same 514x514 padded grid used in filtering, with G=H F. F and G share a log display scale. The box filter has unit DC gain and generally attenuates high frequencies, but unlike a Gaussian response it has zeros and sidelobes rather than a monotonic radial decay.

Are the spatial- and frequency-domain results identical? by the convolution theorem the two methods compute the SAME linear convolution, so the results are identical up to floating-point round-off (MSE ~1e-26, PSNR >300 dB, SSIM=1). The |g_s - g_f| panel confirms it (max ~1e-13).

B-2. Sharpening Filtering

Whole-image mean: input 96.728; sharpening with zero boundaries 97.514.

The sharpening kernel adds a Laplacian-based component to the original. Its coefficients sum to one, giving unit DC gain and preserving constant interior regions. With zero boundaries and same cropping, the mean of the complete finite image can nevertheless change.



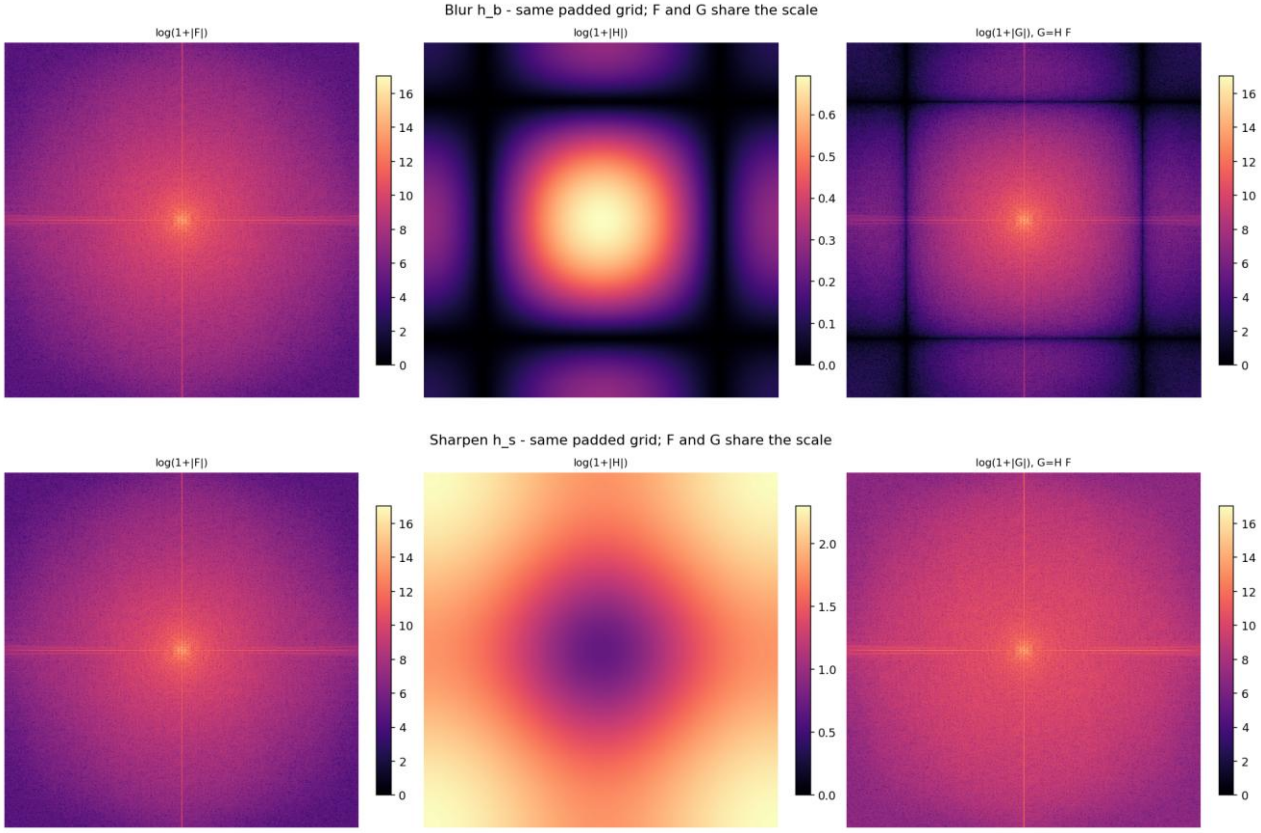
spatial vs frequency: MSE = 4.65e-26, PSNR = 301.5 dB, SSIM = 1.000000

Observed in the figure: coin boundaries and engraved texture become crisper with bright edge halos (overshoot); again the difference map is ~0.

again identical (difference ~round-off). The sharpened output overshoots/undershoots at edges and can leave [0,255], so it is clipped for display only; the metrics use the raw float result.

B-3. Frequency-Domain Analysis

For each filter we compare |F|, |H|, and |G| = |H F|. In the centered spectra the DC/low frequencies are at the middle and the high frequencies toward the edges/corners.



Observed in the figure: the blur |H_b| is bright at the center and dark toward the edges; the sharpen |H_s| is the opposite - dark at the center and bright toward the corners.

Why does the blur filter behave as a low-pass filter? Why blur is low-pass: in centered kernel coordinates, H_b(wx,wy)=(1+2*cos(wx))*(1+2*cos(wy))/9. Its DC gain is one, and high frequencies are attenuated, with zeros and sidelobes. This weakens abrupt intensity changes and fine texture.

Which frequency components does the blur filter attenuate? High-frequency components associated with abrupt intensity changes, edges and fine texture are generally attenuated. The box-filter response has zeros and sidelobes, so attenuation is not monotonic with frequency.

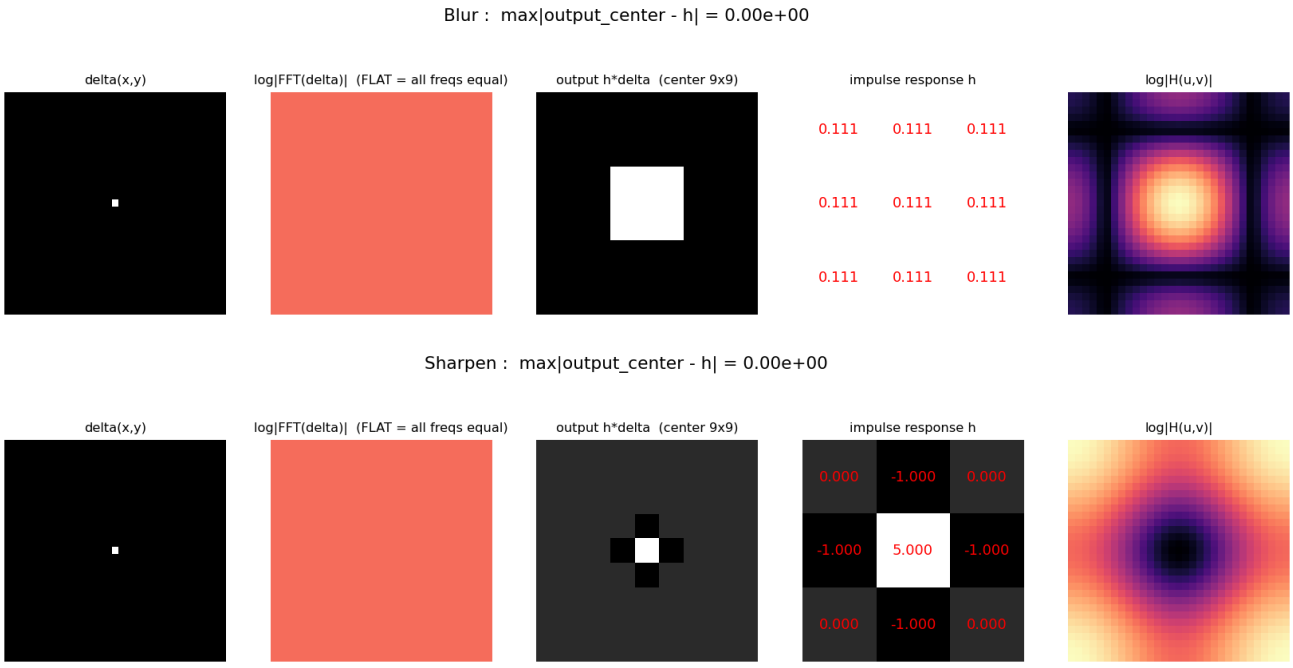
Why does the sharpening filter emphasize high frequencies? H_s(wx,wy)=5-2*cos(wx)-2*cos(wy) has gain one at DC and nine at (pi,pi). It is a high-frequency emphasis filter, not a pure high-pass filter. The dark center of its autoscaled spectrum represents lower relative gain, not zero gain.

How are edges related to high-frequency components? an edge is an abrupt intensity change, which contains strong high-frequency content; so boosting high freq sharpens edges while attenuating it smears them.

B-4. Verification of the Impulse Response

The center of the 31x31 array is the displayed coordinate origin of the impulse. In array-index coordinates this is a shifted impulse, producing the correspondingly shifted h. The shift changes Fourier phase but not its constant unit magnitude. Spatial and FFT impulse outputs are also checked numerically.

Filtering a discrete impulse delta reproduces the impulse response: h*delta = h.



max |output_center - h|: blur = 0.0e+00, sharpen = 0.0e+00

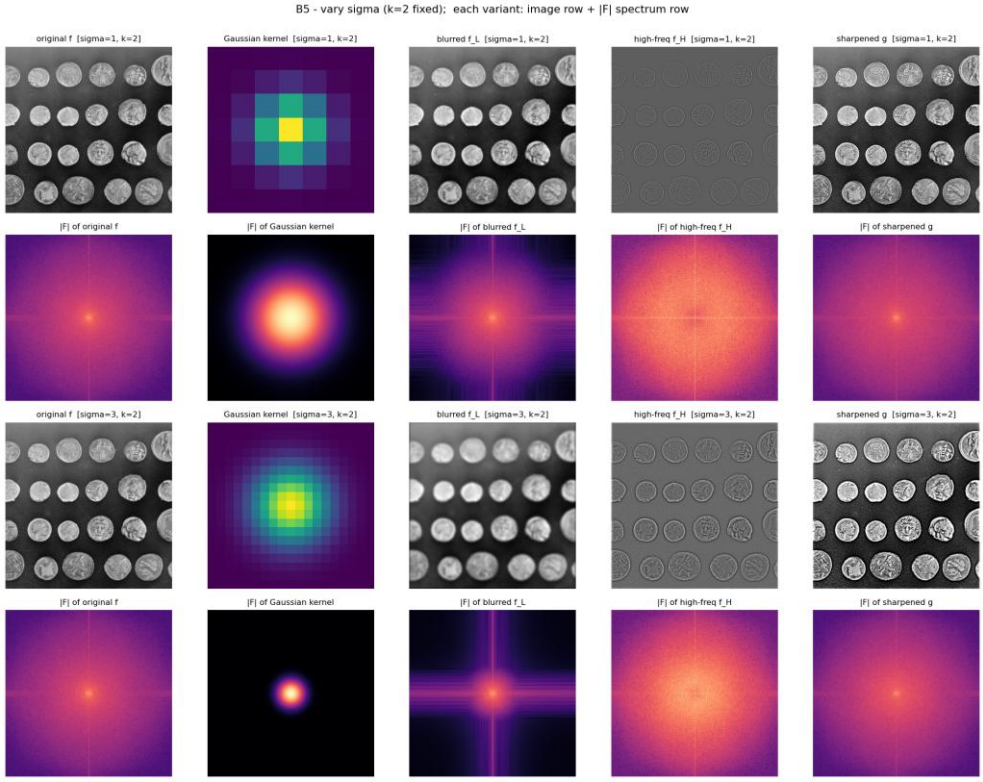
Observed in the figure: the output h*delta (shown as the center 9x9 crop, since h is only 3x3 so the rest of the 31x31 output is zero) is exactly h: a uniform bright block for blur, a bright center with a dark cross for sharpen. The FFT-of-delta panel is one uniform color because |FFT(delta)| = 1 at every frequency (the single color is just the colormap value for that constant).

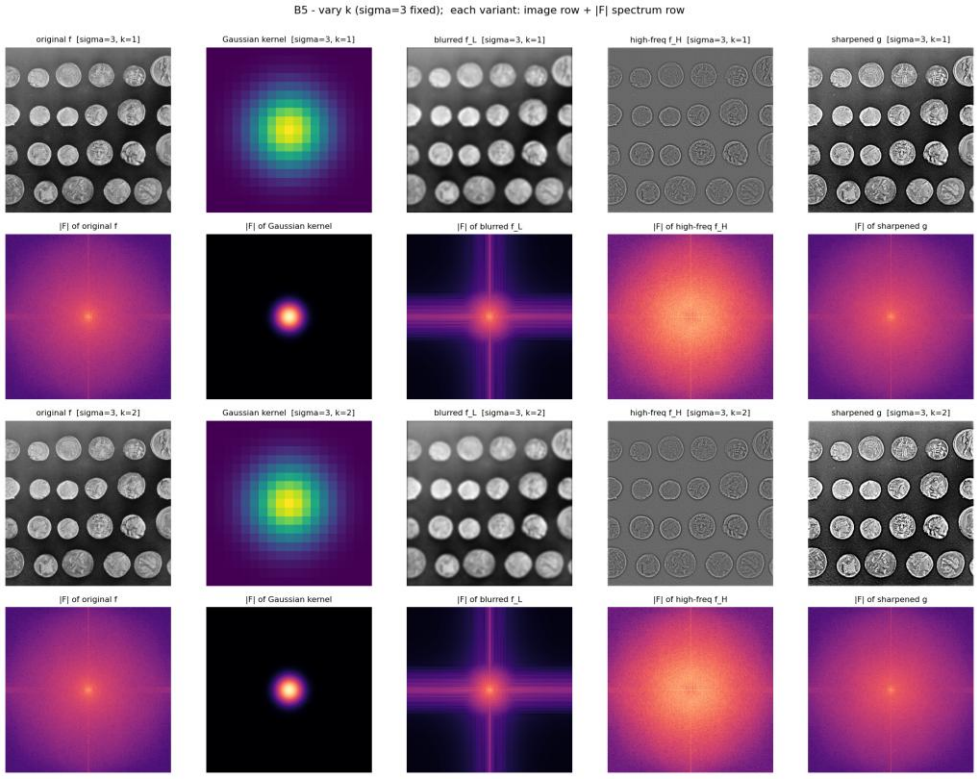
> **Why is h called the impulse response?** feeding the unit impulse delta gives output h*delta = h, so h is literally the system's response to an impulse - hence 'impulse response'. Because delta contains every frequency equally, this one test input reveals the whole LTI system (its spatial response h, equivalently its frequency response H). The measured error is ~0.

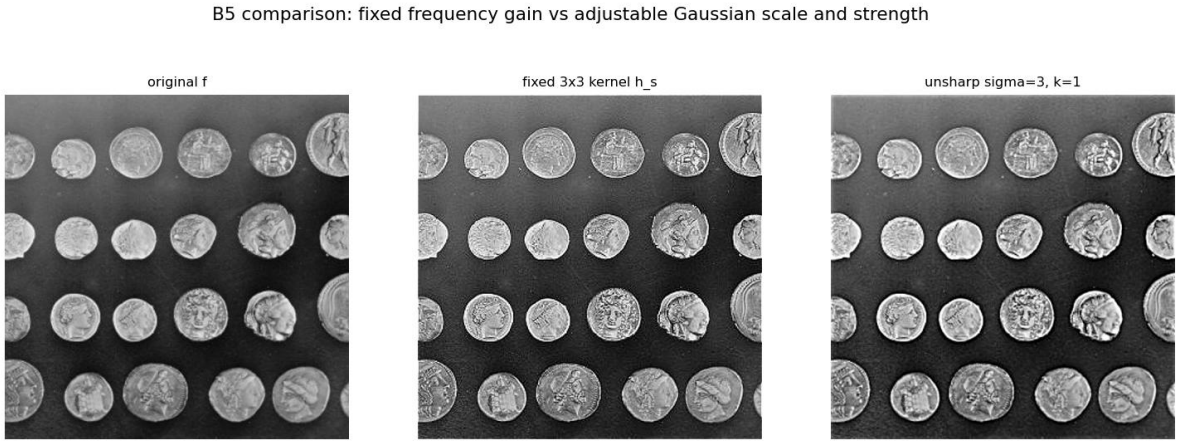
B-5. Gaussian-based Image Sharpening (Unsharp Masking)

Blur with a Gaussian low-pass to get f_L, take the high-frequency part f_H = f - f_L, and add it back: g = (1+k)f - k f_L.

Two 4x5 figures are shown: each variant occupies an image row and a |F| spectrum row; columns are original f, Gaussian kernel, blurred f_L, high-freq f_H, sharpened g. The first figure varies sigma (k fixed), the second varies k (sigma fixed).



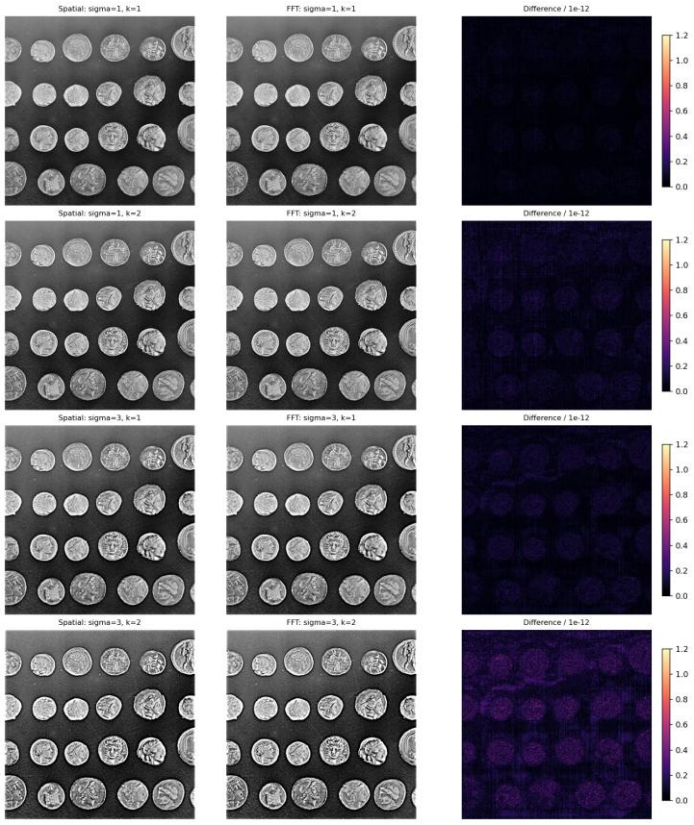




Observed in the figure: increasing sigma broadens the spatial kernel and narrows its low-pass passband. The residual f_H=f-f_L consequently includes a wider frequency range extending toward lower frequencies, producing broader outlines. Increasing k strengthens halos and increases out-of-range output values.

Spatial and frequency verification

The four combinations of sigma=1 or 3 and k=1 or 2 were evaluated in both domains. Gaussian kernel size is 2*ceil(3*sigma)+1, giving 7x7 or 19x19. MSE ranges from 1.84e-27 to 2.52e-26, PSNR from 304.12 to 315.49 dB, and SSIM is one at displayed precision for every pair.



SSIM is one at the displayed precision for every pair, and differences are at round-off scale. At sigma=3, increasing k from 1 to 2 raises the out-of-range fraction from 2.673% to 7.665%, supporting the interpretation of increased saturation and halos with stronger sharpening.

Why does subtracting the Gaussian blur extract high-frequency information? f_L is the low-pass (low-freq) part, so f_H = f - f_L cancels the low freqs and keeps the high freqs (edges/detail) - a Gaussian-based high-pass (visible in the f_H spectrum column).

How does sigma affect the extracted high-frequency components? Effect of sigma: a larger sigma narrows the Gaussian passband. The residual response 1-H_G then emphasizes a range extending to lower frequencies, corresponding to larger spatial structures. Zero boundaries also introduce finite-image boundary effects into the residual.

How does k affect sharpening strength? k scales the high-frequency residual added to the original. Increasing k strengthens edge and detail emphasis.

What happens when k is too large? Excessively large k causes edge overshoot (halos/ringing), noise amplification and more clipping or saturation when displayed.

How does this compare with the fixed sharpening kernel? the 3x3 kernel has fixed frequency-dependent gain. Gaussian unsharp masking allows the spatial scale sigma and strength k to be adjusted separately. The fixed kernel can still respond to broad structures; claiming that it cannot sharpen them at all would be too strong.

B-6. Discussion - Convolution Theorem

$g(x,y) = h(x,y) * f(x,y) \quad <-> \quad G(u,v) = H(u,v)\,F(u,v)$.

Why does spatial convolution correspond to frequency-domain multiplication? write the image as a sum of 2-D sinusoids (its Fourier components). A linear shift-invariant system acts on each sinusoid separately: if the input is a single frequency e^{j2pi(ux+vy)}, the output is the SAME frequency scaled by the complex number H(u,v) (the sinusoid is an eigenfunction of the system). Filtering therefore just multiplies each Fourier component F(u,v) by H(u,v), which is exactly G = H F. So convolution in space corresponds to a per-frequency scaling in the Fourier domain - and for large kernels the FFT route (O(N^2 log N)) is cheaper than direct convolution (O(N^2 k^2)).

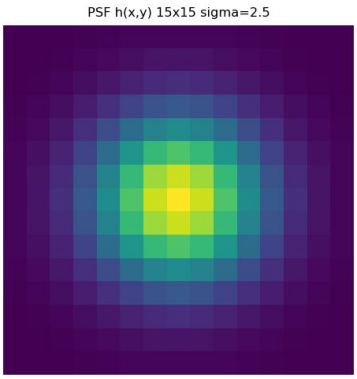
Why can small differences occur in practice? Padding and circular convolution: the FFT product computes circular convolution. Padding both arrays to at least (M+m-1,N+n-1) accommodates the full support of linear convolution, preventing circular overlap. Cropping the inverse transform from the kernel-radius offset aligns the same-size spatial result. Boundary handling: the spatial method assumes a boundary (we used zeros). If the two methods assumed different boundaries (zero vs replicate vs wrap), the border pixels would differ. We used zeros on both, so the borders match. Numerical precision: FFT/IFFT run in finite-precision floating point, so round-off accumulates to ~1e-12..1e-13 even when the math is identical. That is why the measured MSE is ~1e-26 rather than exactly 0.

Part C. Wiener Filter

C-1. Generate a Blurred Image

g_b = h*f, implemented as a frequency-domain multiply (H = FFT of the zero-phase PSF).







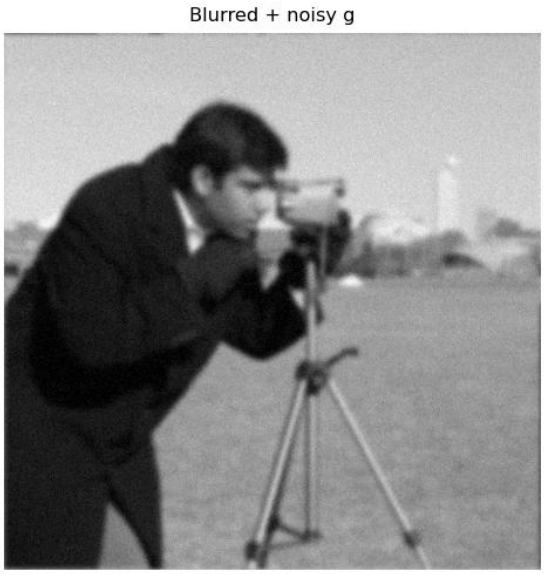
PSF sum = 1.000000 (normalized)

Observed in the figure: the PSF appears as a small bright Gaussian dot, and the blurred cameraman is softened overall, with tripod and background building edges smeared.

C-2. Add Zero-mean Gaussian Noise

g = g_b + n. The generated noise is rescaled so its actual RMS equals the target.



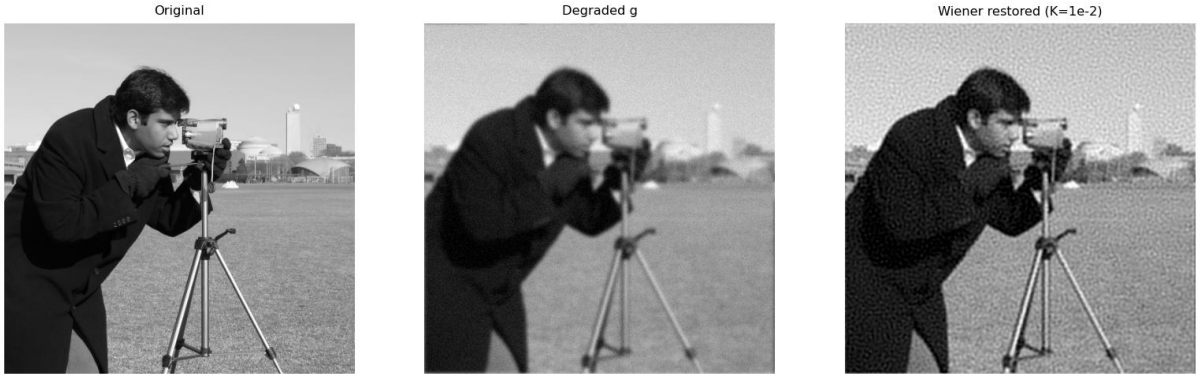


actual noise RMS = 0.03000 (target 0.03000)

Observed in the figure: the right image (blurred+noisy) shows visible grain on the smooth sky/grass, distinguishing it from the left blurred image.

C-3. Restore with a Wiener Filter

The restoration is F_hat=[conj(H)/(|H|^2+K)]*G, followed by the real part of its inverse FFT. K approximates the frequency-dependent noise-to-signal power ratio by a constant. At K=0 the response is 1/H where H is nonzero; inversion is undefined at zeros of H.

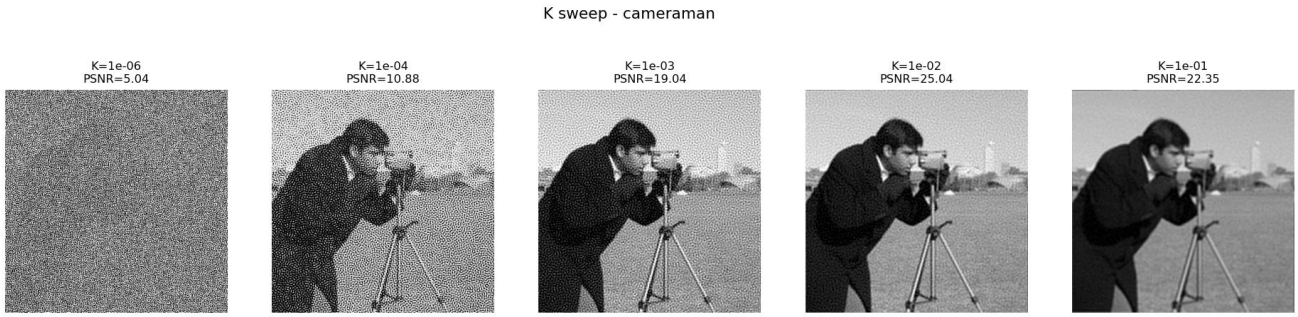


Observed in the figure: the restored (K=1e-2) image removes the blur of the degraded image, sharpening the cameraman/tripod outline and suppressing noise, coming close to the original.

C-4. Effect of K

Evaluation: clip the restored values to [0,1] before comparison with the original. The K table below and the tables in C-5 all use this convention.

K sweep on the primary input (cameraman):



K	MSE	PSNR (dB)	SSIM
1e-06	0.31364	5.04	0.0100
1e-04	0.08169	10.88	0.0577
1e-03	0.01246	19.04	0.2059
1e-02 (best PSNR)	0.00313	25.04	0.5243
1e-01 (best SSIM)	0.00582	22.35	0.6987

Observed in the figure: K=1e-6 severely obscures structure with noise; at 1e-4 the subject is visible amid substantial noise. K=1e-1 is smoother but loses some fine detail.

What happens when K is too small? at frequencies where K is negligible relative to |H|^2, the response approaches 1/H. Undoing attenuation also amplifies noise, especially at small |H|. At K=1e-6 the raw restoration PSNR is -16.37 dB, whereas clipping to [0,1] gives 5.04 dB.

What happens when K is too large? the denominator is dominated by K, so the filter approaches H*/K; deblurring is weak and the restoration is over-smoothed (less noise but lost resolution).

How does K balance noise suppression and sharpness? smaller K allows stronger inversion, whereas larger K suppresses noise amplification. Excessively large K weakens deblurring and fine detail; DC gain is also 1/(1+K), so brightness can decrease. The preferred value depends on the metric, image and noise realization.

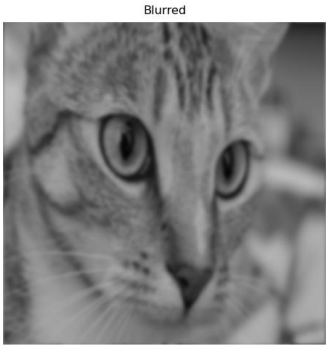
Which K gives the best restoration, and why? Among the five tested values for cameraman, K=1e-2 gives the highest PSNR, balancing noise amplification and excessive smoothing to minimize pixel error. K=1e-1 gives the highest SSIM, so the preferred value depends on the evaluation metric.

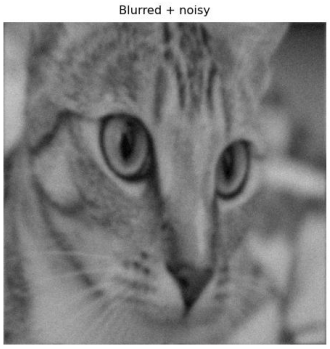
C-5. Results on Additional Inputs

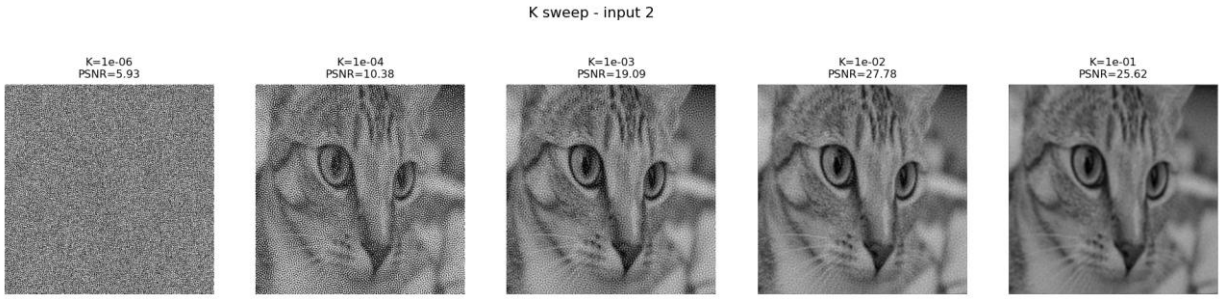
The same degradation + Wiener restoration is repeated on the other input images to check how the optimum K depends on image content.

Input: input 2

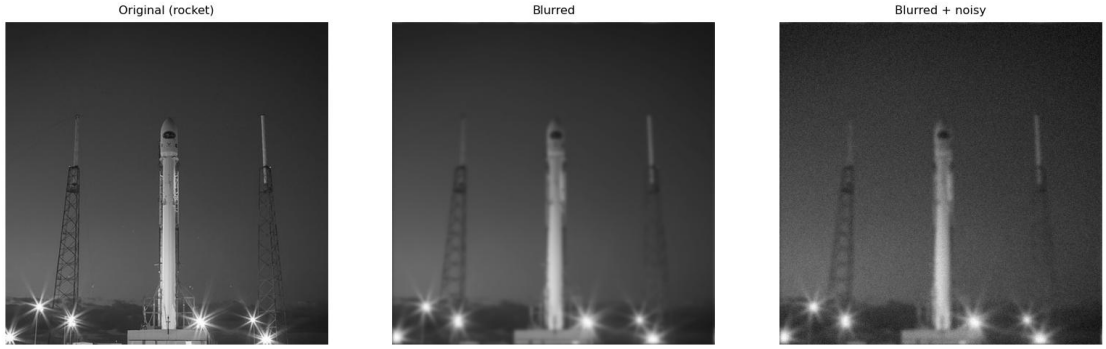


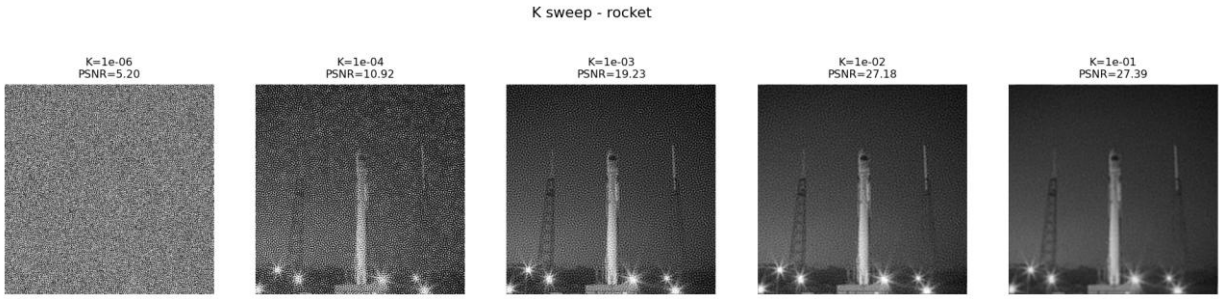






K	MSE	PSNR (dB)	SSIM
1e-06	0.25518	5.93	0.0095
1e-04	0.09165	10.38	0.0374
1e-03	0.01232	19.09	0.1995
1e-02 (best PSNR)	0.00167	27.78	0.6121
1e-01 (best SSIM)	0.00274	25.62	0.8049





K	MSE	PSNR (dB)	SSIM
1e-06	0.30197	5.20	0.0055
1e-04	0.08090	10.92	0.0278
1e-03	0.01195	19.23	0.1418
1e-02	0.00191	27.18	0.5457
1e-01 (best PSNR) (best SSIM)	0.00182	27.39	0.8526

Across these five candidates and the fixed noise realization, cameraman and input 2 have their highest clipped-output PSNR at K=1e-2 and SSIM at K=1e-1. Rocket has both maxima at K=1e-1. These observations demonstrate image- and metric-dependent preferences, not a universal rule for all smooth images.